

Interference-Resilient Optical Wireless Positioning via Machine Learning-Enhanced Subset Filtering

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Abstract *This paper demonstrates a robust machine learning-based optical wireless positioning system that achieves high accuracy under the presence of the interference, enabling reliable position estimation and dynamic interference rejection.*

Introduction

Future wireless networks are envisioned to support a wide range of high-accuracy sensing services, including indoor localization and radar-based environmental perception^[1]. Consequently, high-accuracy indoor positioning has become an indispensable component in modern communication systems. Conventional radio frequency (RF)-based localization methods often suffer from electromagnetic (EM) interference, limiting their applicability in sensitive environments such as healthcare settings^{[2],[3]}.

Optical wireless positioning (OWP) has emerged as a promising alternative due to its superior accuracy, achieving centimeter-level positioning errors (e.g., below 10 cm) and offering greater energy efficiency and EM immunity compared to Wi-Fi, Bluetooth, and ultra-wideband (UWB) systems^{[4],[5]}.

However, existing OWP solutions typically assume interference-free conditions, often ignoring the impact of external interference sources. In practice, spectral overlaps between positioning subcarriers and interference signals can severely degrade system performance^[6].

To address these challenges, we propose a machine learning (ML)-based OWP framework that mitigates interference, recovers degraded position estimates, and detects the interference sources. Our design introduces the concept of subset-based received signal strength indicator (RSSI) inputs. We further develop and experimentally validate a reliability-aware subset fusion mechanism that detects anomalies, excludes corrupted signals, and aggregates reliable estimations to restore positioning accuracy. Through this approach, the proposed system remains robust and accurate even in the presence of spectral interference.

OWP Channel Model

To support simultaneous optical wireless communication (OWC) and OWP, we consider employing

the downlink channel for data transmission, while utilizing the uplink channel for positioning and communication. Specifically, the indoor positioning environment is equipped with N_r ceiling-mounted angle diversity receivers (ADRs) facing downward, whose novel design ensures that, regardless of the user equipment (UE) position, at least one photodetector (PD) surface is oriented toward it^[7]. The user carries an infrared (IR)-LED that transmits an N_r -carrier on-off keying-frequency division multiplexing (OOK-FDM) signal. Each receiver is assigned to a distinct subcarrier and independently records and stores the RSSI corresponding to that subcarrier. The total channel gain between the IR-LED and the i -th ADR is calculated^{[7],[8]}, based on which, the signal-to-noise ratio (SNR) at each receiver is derived. This SNR serves as the RSSI, which is then used for subsequent positioning tasks.

Interference-Resilient OWP Subset Generation

In conventional ML-based OWP systems, the RSSI at each receiver is directly used to train neural networks for position estimation^[8]. However, in the proposed interference-aware uplink scenario, external sources emit signals that partially overlap with the OOK-FDM subcarriers, contaminating the observed spectral environment and leading to substantial degradation in positioning accuracy.

To address this, we introduce the concept of RSSI subsets. The system does not require prior knowledge of the number or locations of interferers but instead generates all possible combinations of RSSI indices, with subset sizes varying from a minimum threshold—ensuring sufficient positioning information—to the full set. Each subset of RSSI indices is denoted by ρ_k , where $k = 1, 2, \dots, K$, and K represents the total number of subsets generated. Since the UE orientation has a significant impact on positioning accuracy^[4], in the dataset, the subset vectors ρ_k and the corresponding orientation angles are used as inputs, while the 3D position (x, y, z) serves as the output label for tar-

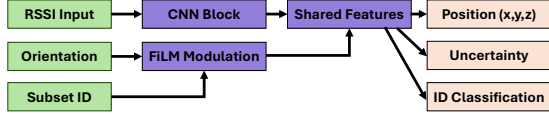


Fig. 1: Hybrid CNN-MLP architecture

get UE positioning.

Hybrid Convolutional Neural Network (CNN)-Multilayer Perceptron (MLP) Architecture

In this work, we propose a multi-task localization framework that integrates Feature-wise Linear Modulation (FiLM) and subset-based fingerprinting with machine learning to enhance robustness against interference in OWP systems, as shown in Fig. 1. Specifically, the model comprises three input modalities: RSSI subsets, UE orientation information, and cluster indices indicating different subsets of RSSI. Then, the final estimated position is given by $\hat{\mathbf{p}}_k$ through the network. We further quantify the positioning uncertainty of each subset ρ_k by minimizing a residual loss defined over the set of receivers. For an estimated position $\hat{\mathbf{p}}_k$ corresponding to the k -th RSSI subset, we compute the weighted least squares residual with respect to the known positions $\mathbf{p}_{\text{ADR}}^{(i)}$ of the receivers:

$$\hat{\sigma}_k = \min_{\hat{\mathbf{p}}_k} \sum_i \left(\frac{\|\mathbf{p}_{\text{ADR}}^{(i)} - \hat{\mathbf{p}}_k\|_2}{\text{RSSI}_i} \right)^2, \quad (1)$$

where RSSI_i denotes the received signal strength at the i -th receiver. This formulation treats RSSI as a reliability weight, penalizing larger deviations between the estimated position and the receiver locations more heavily when the signal strength is high. The resulting residual $\hat{\sigma}_k$ serves as an estimate of the uncertainty associated with the $\hat{\mathbf{p}}_k$.

Robust Position Estimation through Interference-Aware Subset Filtering

To consolidate the estimations from multiple RSSI subsets under interference, each subset output pair $(\hat{\mathbf{p}}_k, \hat{\sigma}_k)$ is treated as a Gaussian-distributed localization hypothesis. A temporary estimate $\tilde{\mathbf{p}}$ is computed via uncertainty-aware fusion^[9]:

$$\tilde{\mathbf{p}} = \left(\sum_{k=1}^K \hat{\mathbf{p}}_k \oslash \hat{\sigma}_k \right) \oslash \left(\sum_{k=1}^K \mathbf{1} \oslash \hat{\sigma}_k \right), \quad (2)$$

where $\oslash$ denotes the Hadamard (element-wise) division. Each subset's consistency is evaluated by the deviation metric^[9]:

$$d_k = \|\hat{\mathbf{p}}_k - \tilde{\mathbf{p}}\|_2, \quad (3)$$

Subsets exhibiting excessive deviation are considered unreliable and are excluded from the final fusion. Specifically, a subset l is flagged if its deviation exceeds a dynamic threshold Λ , given by^[9]:

$$\Lambda = \mathbb{E}[d_k] + n_\Lambda \sqrt{\mathbb{V}[d_k]}, \quad (4)$$

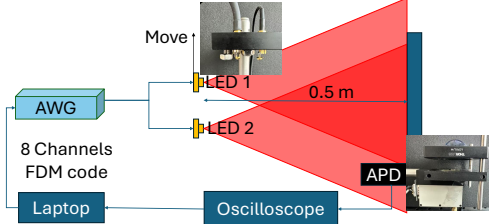


Fig. 2: Experiment set-up for ML-based OWP with the interference

where n_Λ is a tunable confidence parameter, typically set to 3 based on the three-sigma rule. This filtering mechanism ensures that only statistically consistent subset estimations contribute to the final estimate. The final position estimate, denoted as $\mathbf{p}_{\text{final}}$, is computed as the arithmetic mean of the retained, validated subset estimations^[9]:

$$\mathbf{p}_{\text{final}} = \frac{1}{|\mathcal{L}_{\text{valid}}|} \sum_{l \in \mathcal{L}_{\text{valid}}} \hat{\mathbf{p}}_l, \quad (5)$$

where $\mathcal{L}_{\text{valid}}$ denotes the set of subset indices that passed the outlier rejection test. After evaluating the deviation of each subset estimation from the fused estimate, the indices of non-benign subsets are collected in the set $\mathcal{L}_{\text{out}}$. To identify the interfering sources, we perform a voting-based exclusion mechanism. Specifically, the identifiers of potentially corrupted sources are obtained by intersecting the channel indices across all non-benign subsets^[9]:

$$\bigcap_{l \in \mathcal{L}_{\text{out}}} \rho_k, \quad (6)$$

To account for potential estimation noise, we implement a majority-vote refinement strategy. Let j denote a particular RSSI channel. Then, we define^[9]:

$$A = \sum_{l \in \mathcal{L}_{\text{out}}} \mathbb{I}\{j \in \rho_k\}, \quad B = \sum_{l \notin \mathcal{L}_{\text{out}}} \mathbb{I}\{j \in \rho_k\}, \quad (7)$$

If $A > B$, the channel j is considered to be concurrently observed by both the interference source and the target user, indicating that the corresponding subcarrier frequency is contaminated and should be excluded from subsequent positioning computation.

Experiment Set-up

To prove the concept, an experimental system was implemented to validate the proposed ML-assisted interference mitigation and robust position estimation approach for uplink Near-IR (NIR)-LED-based OWP. The experimental setup is illustrated in Fig. 2. The optical transmitter consists of an NIR-LED (LED1) representing the target user, placed at a transmission distance of 0.5 m and positioned randomly within the environment. LED1 is driven by an 8-carrier OOK-FDM waveform (300 mV_{pp}) generated by an Arbitrary Waveform Generator (AWG), superimposed on a 50 mA

DC bias supplied externally. On the receiver side, an avalanche photodiode (APD) coupled with a condenser lens is mounted on a motorized translation stage and aligned to face the LED1, simulating the scenario in which a PD surface of an ADR is oriented toward the transmitter. The detector is sequentially positioned at eight predefined locations. At each position, the analog output from the APD is digitized using a high-speed oscilloscope. The fast fourier transform is then applied to extract the RSSI values across all refined subcarriers for model training. To emulate an interference-aware environment, a secondary FDM signal is generated by another AWG channel and transmitted through a second NIR-LED (LED2). This signal partially overlaps with an arbitrary subcarrier from LED1, introducing the interference similar to that caused by infrared sensors or positioning tags operating in environments such as hospitals, factories, or large commercial spaces, where concurrent transmissions can unintentionally contaminate the frequency bands used for positioning.

Results

To validate the performance of the proposed method under practical conditions, we first construct a MATLAB-based simulated positioning system that closely replicates the experimental setup, and use it to generate training data for the model. In this setting, 1000 test samples were generated in MATLAB to evaluate the system, resulting in an average Euclidean positioning error of only 0.947 cm. Meanwhile, real-world experiments were conducted, as shown in Fig. 3, the estimated positions exhibit minimal deviation from the ground truth positions after applying the proposed method. In contrast, the baseline positioning system^[4] experiences substantial degradation when interference sources are present. In addition, the proposed interference identification mechanism successfully detects overlapping spectral channels between the interference sources and the target user with a success rate of 99.3% in the 1000 test samples. Furthermore, in practical experiments, the system consistently identifies the overlapping channels in every test. The validation is performed by comparing the interference channels identified by the proposed method with the known injected interference frequencies, confirming the consistency between the estimated and actual affected subcarriers, which further demonstrating the feasibility and robustness of the proposed approach. We have shown experimentally that the proposed method works for a small-scale test environment. To further validate the scalability and robustness of the proposed framework, simulations are conducted for a larger indoor environment with dimensions of $5 \times 5 \times 3$ meters. In this scenario, the UE is randomly positioned and randomly ori-

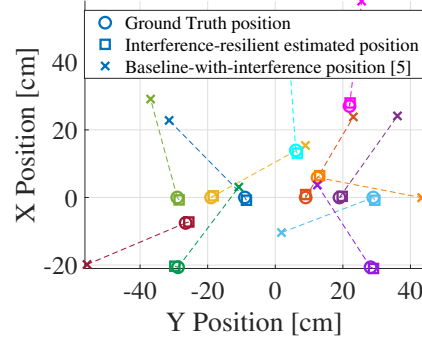


Fig. 3: The experiment validation for the proposed model

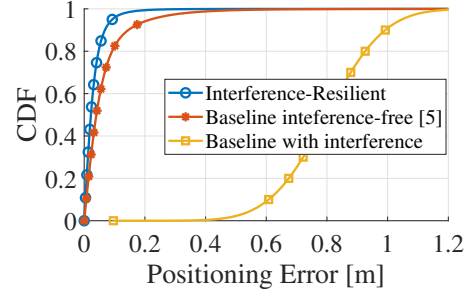


Fig. 4: Simulation for positioning error in CDF

ented in the indoor environment. As illustrated in Fig. 4, the cumulative distribution function (CDF) of the positioning error clearly demonstrates the advantage of the proposed interference-resilient model. Compared to the baseline interference-free system^[4], the proposed approach achieves significantly lower positioning errors, with over 90% of the test cases exhibiting errors below 10 cm, even under random orientation conditions. In contrast, the baseline system exhibits significantly higher positioning errors under interference conditions, leading to substantial performance degradation, with the majority of errors exceeding 0.6 m. These results confirm that the proposed method maintains high positioning accuracy and strong robustness against environmental uncertainties and spectral interference. Moreover, the proposed model achieves a mean Euclidean positioning error of only 5 cm, representing a substantial improvement even over the baseline system under interference-free conditions. This improvement is attributed to the subset filtering mechanism, which aggregates multiple reliable subsets rather than relying on a single biased estimation.

Conclusion

We proposed a machine learning-enhanced OWP framework that achieves robust positioning under spectral interference through subset filtering and reliability-aware fusion. Experimental results demonstrate that the proposed method achieves centimeter-level positioning accuracy and reliably identifies interfering spectral channels. Future work will extend experimental validation to larger environments and will address more complex interference case.

Acknowledgements

This work was supported by the Future Telecoms Research Hub, Platform for Driving Ultimate Connectivity (TITAN), sponsored by the Department of Science Innovation and Technology (DSIT) and the Engineering and Physical Sciences Research Council (EPSRC) under Grant EP/X04047X/1 and Grant EP/Y037243/1

References

- [1] A. Liu, Z. Huang, M. Li, *et al.*, “A survey on fundamental limits of integrated sensing and communication”, *IEEE Communications Surveys and Tutorials*, vol. 24, no. 2, pp. 994–1034, 2022. DOI: 10.1109/COMST.2022.3149272.
- [2] J. Wyffels, J.-P. Goemaere, P. Verhoeve, P. Crombez, B. Nauwelaers, and L. De Strycker, “A novel indoor localization system for healthcare environments”, in *2012 25th IEEE International Symposium on Computer-Based Medical Systems (CBMS)*, 2012, pp. 1–6. DOI: 10.1109/CBMS.2012.6266347.
- [3] T. Koonen, “Indoor optical wireless systems: Technology, trends, and applications”, *Journal of Lightwave Technology*, vol. 36, no. 8, pp. 1459–1467, 2018. DOI: 10.1109/JLT.2017.2787614.
- [4] Y. Huang, M. Safari, H. Haas, and I. Tavakkolnia, “Optical wireless 3-d-positioning and device orientation estimation”, *IEEE Open Journal of the Communications Society*, vol. 5, pp. 4519–4530, 2024. DOI: 10.1109/OJCOMS.2024.3423420.
- [5] S. Bastiaens, J. V.-V. Gerwen, N. Macoir, *et al.*, “Experimental benchmarking of next-gen indoor positioning technologies (unmodulated) visible light positioning and ultra-wideband”, *IEEE Internet of Things Journal*, vol. 9, no. 18, pp. 17 858–17 870, 2022. DOI: 10.1109/JIOT.2022.3161791.
- [6] A. Costanzo, V. Loscri, V. Deniau, and J. Rioult, “On the interference immunity of visible light communication (vlc)”, in *GLOBECOM 2020 - 2020 IEEE Global Communications Conference*, 2020, pp. 1–6. DOI: 10.1109/GLOBECOM42002.2020.9348209.
- [7] Z. Zeng, M. D. Soltani, M. Safari, and H. Haas, “Angle diversity receiver in lifi cellular networks”, in *ICC 2019 - 2019 IEEE International Conference on Communications (ICC)*, 2019, pp. 1–6. DOI: 10.1109/ICC.2019.8761200.
- [8] M. A. Arfaoui, M. D. Soltani, I. Tavakkolnia, *et al.*, “Invoking deep learning for joint estimation of indoor lifi user position and orientation”, *IEEE Journal on Selected Areas in Communications*, vol. 39, no. 9, pp. 2890–2905, 2021. DOI: 10.1109/JSAC.2021.3064637.
- [9] W. Liu and P. Papadimitratos, “Position-based rogue access point detection”, in *2024 IEEE European Symposium on Security and Privacy Workshops (EuroS&PW)*, 2024, pp. 436–442. DOI: 10.1109/EuroSPW61312.2024.00055.